

Graph metrics

Sub-characteristics

Composite

Diagnosis → remedy

Reverse PageRank,
in-degree, cascade
depth on G^T

Directed articulation,
QoS-SPOF, bridges,
CDI, QoS weight

Betweenness, QoS fan-
out, code quality,
coupling, clustering

Fault Tolerance FT

$$0.45 \text{ RPR} + 0.30 \text{ Deg}_{\text{in}} \\ + 0.25 \text{ CDPot}$$

Availability A

$$0.25 \text{ AP} + 0.20 \text{ QSPOF} + 0.20 \text{ BR} \\ + 0.25 \text{ CDI} + 0.10 w$$

Maintainability M

$$0.35 \text{ BT} + 0.30 w_{\text{out}} + 0.15 \text{ CQP} \\ + 0.12 \text{ CR} + 0.08 (1 - \text{CC})$$

Reliability R

$$0.36 \text{ FT} + 0.64 \text{ A}$$

Quality Q(v)

$$0.80 \text{ R} + 0.20 \text{ M}$$

Tukey fence →
CRITICAL tier

High A, low FT

single point of failure
→ replication

High FT

error-cascade hub
→ circuit breaker

High M

coupling bottleneck
→ decoupling, refactoring

each flagged component is read
by its FT / A / M profile